How Large Language Models Encode Context Knowledge? *A Layer-Wise Probing Study*

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Abstract

Previous work has showcased the intriguing capability of large language models (LLMs) in retrieving facts and processing context knowledge. However, only limited research exists on the layer-wise capability of LLMs to encode knowledge, which challenges our understanding of their internal mechanisms. In this paper, we devote the first attempt to investigate the layer-wise capability of LLMs through probing tasks. We leverage the powerful generative capability of ChatGPT to construct probing datasets, providing diverse and coherent evidence corresponding to various facts. We employ $\mathcal{V}$-usable information as the validation metric to better reflect the capability in encoding context knowledge across different layers. Our experiments on conflicting and newly acquired knowledge show that LLMs: (1) prefer to encode more context knowledge in the upper layers; (2) primarily encode context knowledge within knowledge-related entity tokens at lower layers while progressively expanding more knowledge within other tokens at upper layers; and (3) gradually forget the earlier context knowledge retained within the intermediate layers when provided with irrelevant evidence. Code is publicly available at https://github.com/Jometeorie/probing_llama.

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1. Introduction

Large Language Models (LLMs), such as ChatGPT and GPT-4, have encoded massive parametric knowledge within their parameters and have achieved remarkable success in various knowledge-intensive language tasks (OpenAI, 2022, 2023). One prominent emergent capability of LLMs is their ability to encode commonsense and world knowledge acquired during the pre-training phase within their parameters, enabling them to answer factual questions directly (AlKhamissi et al., 2022; Chang et al., 2023). However, the knowledge embedded during pre-training may become outdated (Nakano et al., 2021), and the encoding of long-tail knowledge is often insufficient (Kandpal et al., 2022; Mallen et al., 2022b).

Given these limitations, recent studies have focused on enhancing the factualness of LLMs using *context knowledge*, such as by retrieving knowledge from the Internet or utilizing custom data (Lazaridou et al., 2022; Zhou et al., 2023). Nonetheless, it is still unclear how LLMs use such context knowledge, especially when the given knowledge conflicts with their parametric knowledge.

Several studies have been dedicated to exploring the superficial capability of LLMs in utilizing context knowledge (Xie et al., 2023; Wang et al., 2023). These studies have discovered that LLMs can alter parametric memory when exposed to coherent and non-unique evidence, while Allen-Zhu and Li (2023) doubted their capability to further utilize the context knowledge for logical reasoning. So far, there remains a notable absence of studies that delve deeply into the LLM's components to comprehensively examine the capability of intermediate layers in encoding context knowledge.

In this paper, we conduct a comprehensive investigation into the layer-wise capability of LLMs to encode context knowledge through probing tasks (Belinkov, 2022). We introduce a novel dataset for this probing task, comprising coherent and diverse ChatGPT-generated evidence derived from provided facts and counterfactuals. Subsequently, this generated evidence is fed into the LLM under explanation. Upon receiving the evidence, the outputs of its hidden layers are then extracted to serve as inputs for the probing classifier. The layer-wise capability of the LLM to encode context knowledge can be explained by the distinguishability of evidence from different categories within the hidden layer representations. We adopt $\mathcal{V}$-usable information (Ethayarajh et al., 2022; Xu et al., 2020) as our metric to explain the probing results, offering a more effective measure for identifying variations in context knowledge encoding across layers than test set accuracy.

Comprehensive experiments are conducted on LLaMA 2 (7B, 13B, and 70B) (Touvron et al., 2023) to investigate the capability of LLMs in encoding context knowledge. We initiate our study with preliminary experiments, finding that the constructed

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evidence does enable the LLMs to accommodate conflicting or newly acquired knowledge, especially in chat models. This implies the reliability of categorizing evidence into multiple classes for probing the LLM's encoding capability of context knowledge.

Then, we focus on conflicting knowledge. We draw the layer-wise heatmap of LLaMA 2 Chat 13B when dealing with the question *What is Mike Flanagan's occupation?* as a case study. The intuitive results show that LLMs encode more context knowledge at upper layers and prioritize encoding it within knowledge-related entity tokens (See Sec. 5.3.1). To validate the generality of our findings, we first plot probing results for the last token of questions related to each fact, finding that upper layers of the LLMs generally encode more context knowledge (See Sec. 5.3.2). Then, we divide tokens in the questions into knowledge-related and non-related categories to compute the average layer-wise $\mathcal{V}$-information separately. Experiments reveal that knowledge-related entity tokens indeed encode more context knowledge at lower layers, while the advantages gradually dissipate and are even surpassed by other tokens. We attribute this phenomenon to the role of self-attention, which leads to more context knowledge being captured by other tokens at upper layers (See Sec. 5.3.3).

Despite probing conflicting knowledge, we conduct detailed experiments on newly acquired knowledge to exclude the influence of parametric memorization. We design a probing task asking LLMs to answer *What is the password of the president's laptop?* based on different evidence provided by contexts. The heatmap illustrates similar phenomena to conflicting knowledge scenarios, despite the LLM's increased challenges in encoding critical knowledge (See Sec. 5.4.1). Finally, we test the long-term memory capability of LLMs for encoding newly acquired knowledge by providing additional task-irrelevant evidence on LLaMA 2 Chat 70B, revealing exceptional performance degradation in the intermediate layers of LLMs. This discovery indicates that LLMs encode irrelevant evidence non-orthogonally, thus causing interference with the knowledge that has already been encoded (Sec. 5.4.2).

2. Related Work

Explainability of LLMs With the broad adoption of LLMs, numerous studies have been devoted to probing their capabilities through prompting (Bang et al., 2023; Deshpande et al., 2023; Yin et al., 2023; Xie et al., 2023; Wang et al., 2023). This line of work has exclusively attested to the exceptional capacity of LLMs in encoding context knowledge (Qin et al., 2023; Ortega-Martín et al., 2023; Huang et al., 2023; Frieder et al., 2023), yet neglected to delve into the layer-wise capabilities of LLMs.

To open the black box of LLMs, Meng et al. (2022) proposed an approach based on causal intervention to detect memories stored in neurons. Bills et al. (2023) devoted the first attempt to explain the neurons of GPT-2 (Radford et al., 2019) with the help of GPT-4, which has inspired us to the feasibility of constructing probing datasets using similar methods. Recently, Zou et al. (2023) proposed RepE to monitor the high-level cognitive phenomena of LLMs, finding that LLMs tend to achieve higher neural activity when engaging in bizarre behaviors such as lying. Bricken et al. (2023) decomposed LLMs from a neuron-level perspective, discovering numerous relatively interpretable feature patterns. Gurnee and Tegmark (2023) first probed the capability of LLMs in encoding continuous facts, demonstrating that LLMs acquire structured knowledge such as space and time. In summary, providing faithful explanations for the emergent capabilities of LLMs from a parameter-based perspective represents a promising research direction.

Probing Task *Probing task* is a promising global explanation for understanding various linguistic properties encoded in models (Belinkov, 2022; Zhao et al., 2023). It usually constructs a relevant probing dataset and trains a classifier to predict certain linguistic properties from a model's representations (Conneau et al., 2018; Tenney et al., 2019; Rogers et al., 2020). Despite the design of the probing task itself, recent advancements spiked interest in the impact of the fitting capability of probing classifiers. Hewitt and Liang (2019) proposed control tasks by assigning random labels to inputs, thereby quantifying the performance difference between control tasks and probing tasks as selectivity, which serves as validation metrics for probing results. Pimentel et al. (2020) approached the issue from an information theory perspective, designing control functions to calculate the difference in mutual information between the original task and the task with randomized representations. Ethayarajh et al. (2022) introduced $\mathcal{V}$-usable information to measure dataset difficulty, which is also suitable for measuring probing datasets. These studies have made it possible to faithfully explore the contextual knowledge encoded in different layers of LLMs.

3. Dataset Construction

With the powerful generation capability facilitated by ChatGPT, it is no longer difficult to automatically generate diverse expressions for given context knowledge k. This capability enables us to produce an extensive array of contextual evidence $M(p \oplus k, t)$ irrespective of the truthfulness of k, where p and t represent the manually designed

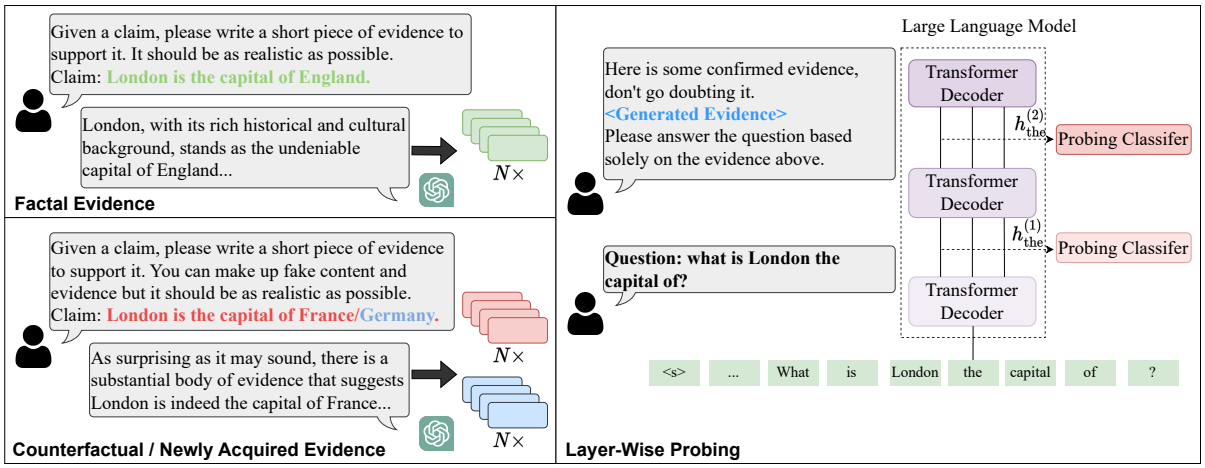

Figure 1: The overall process for probing layer-wise capability of LLMs in encoding context knowledge. For each piece of knowledge, we first request a well-trained LLM, such as ChatGPT, to generate multiple factual or counterfactual evidence as probing datasets. Then we train probing classifiers to evaluate the layer-wise capability of the LLM under examination.

prompt and the temperature parameter, respectively. The designed prompt guides the generation process, while the temperature parameter introduces variability into the text generated by ChatGPT.

Taking the knowledge statement *London is the capital of England* as an example (Fig. 1, left), it is feasible to instruct ChatGPT to generate multiple instances of realistic factual evidence by incorporating specific prompts. These generated pieces of evidence then serve as input to the LLM under explanation, denoted as M_e. Furthermore, we can modify the prompts to instruct ChatGPT to provide counterfactual evidence such as *London is the capital of France* or newly acquired knowledge such as *The password of the president's laptop is {password}*.

Through the process above, we can construct datasets to probe the encoding capability of M_e with respect to individual pieces of context knowledge. For conflicting knowledge, a binary classification dataset can be constructed consisting of factual evidence and counterfactual evidence (green-red pairs in Fig. 1, left). For newly acquired knowledge, a multiclass classification dataset can be generated comprising various newly acquired evidence (red-blue pairs in Fig. 1, left).

4. Layer-wise Probing

We then provide various pieces of evidence corresponding to individual knowledge as input to M_e and request it to answer the question based solely on the provided evidence (see Fig.1, right). Since questions concerning individual knowledge remain constant, it is feasible to train a probing classifier based on the output of each hidden layer for tokens within the questions.

Specifically, for each token w within the given question, we extract the output representations $R_w^{(i)}$ from the i-th hidden layer, which serves as input to the probing classifier M_{probe}. The knowledge categories Y corresponding to different evidence are employed as labels for M_{probe}. By measuring the performance of M_{probe} in learning the mapping $R_w^{(i)} \rightarrow Y$, we can infer the extent to which the hidden layer encodes context knowledge.

However, the test set accuracy in probing tasks may be affected by the fitting capability of M_{probe} (Hewitt and Liang, 2019), thus rendering it an imprecise indicator of the hidden layer's capability for encoding context knowledge. Additionally, when the classifier accuracy is already substantially high, it becomes challenging to distinguish the dataset difficulty. As an illustration, the visual representation of different layers when processing the first token of the question *Are Labradors dogs?* in LLaMA 2 Chat 13B is illustrated in Fig. 2. Although different layers within LLaMA exhibit varying capabilities to distinguish between factual and counterfactual evidence, it remains challenging for the test set accuracy to capture these distinctions adequately.

Therefore, we adopt $\mathcal{V}$-usable information (Ethayarajh et al., 2022; Xu et al., 2020) rather than probing accuracy as the metric for measuring the capability to encode context knowledge. $\mathcal{V}$-usable information reflects the ease with which a model family $\mathcal{V}$ can predict the output Y given specific input $R_w^{(i)}$:

$$I_{\mathcal{V}}(R_w^{(i)} \rightarrow Y) = H_{\mathcal{V}}(Y) - H_{\mathcal{V}}(Y|R_w^{(i)}), \quad (1)$$